THALLAPALLI INDRASENA REDDY

AI / ML Engineer

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PROFILE SUMMARY

AI/ML Engineer with hands-on experience building enterprise GenAI applications, RAG-based assistants, agentic workflows, workforce intelligence solutions, and AI-enabled data pipelines. Skilled in LangGraph, LLM orchestration, vector retrieval, MongoDB, backend APIs, and cloud infrastructure, with a focus on integrating AI systems with enterprise data, business rules, validation layers, and production workflows.

WORK EXPERIENCE

AI / ML Engineer — Onusworks **June 2025 – Present**

Bangalore, India (On-site)

- Built enterprise AI/GenAI systems across workforce analytics, employee support, scheduling, recruitment, and data-intelligence use cases by combining LLM workflows with backend services and enterprise data.
- Developed LangGraph- and RAG-based workflows for VYN, including sliding-window conversational memory, natural-language-to-MongoDB query generation, vector retrieval, schema verification before execution, and secure multi-tenant data access.
- Developed the Crew Call engine layer for AI-assisted workforce scheduling, combining employee eligibility, availability and preference signals, historical assignment context, business-rule validation, and enterprise workforce-management scheduling APIs.
- Built an AI-assisted EDW report rationalization workflow to classify legacy reports, map equivalent reports to BI dashboards or ERP transaction codes, and trace new or enhancement cases to supporting source evidence while reducing manual cross-file analysis.
- Built backend and cloud integrations across AI workflows, including face-verification APIs, REST services, and GCP/Terraform infrastructure for enterprise applications.
- Worked on SG&A expense anomaly detection to identify abnormal spending and potential cost leakage, and on OTIF prediction to estimate whether future orders would be delivered on time and in full from historical logistics data; later evaluated Snowflake-native AI/ML capabilities for combined live and historical analytics.

Machine Learning Engineer (Intern) — VCodez **Feb 2025 – May 2025**

Chennai, India

- Developed ML workflows using Python, Scikit-learn, and Pandas, covering data preprocessing, feature engineering, model training, and evaluation.
- Built an AI-powered outfit suggestion prototype using ML-based image recognition and collaborated with application teams to integrate model outputs into real-time workflows.
- Created stakeholder-facing visualizations in Power BI and Excel and mentored junior interns on Python, machine learning fundamentals, and AI tooling.

PROJECTS

MAJOR PROJECTS

VYN Insights – Enterprise AI Assistant

Node.js · React · MongoDB · LangChain · OpenAI · LangGraph · Zod · Chart.js · Ant Design · RAG · OpenRouter

- Built an analytics agent that translates natural-language questions into MongoDB Aggregation Pipelines (MQL) for enterprise workforce and compliance data.
- Implemented a schema-aware RAG flow that discovers relevant collections, retrieves schema context with MongoDB Vector Search and HyDE-assisted retrieval, verifies fields against source schemas, and validates generated MQL with Zod before execution.
- Added sliding-window conversational memory for follow-up query chaining, plus tenant isolation, projection controls, PII guardrails, and adaptive response rendering as summaries, tables, or charts.

Vinny – Internal AI Assistant

Node.js · LangChain · MongoDB Vector Search · OpenAI Embeddings · Gemini · JavaScript · RAG · LLMs

- Built a context-aware RAG assistant for Crew and Employee Management workflows with screen-aware context so retrieval and guidance adapt to the user's current application location.
- Designed a three-layer cache covering embedding reuse, intent-level reuse, and semantic-similarity response reuse to reduce repeated LLM calls and improve response latency.
- Implemented deterministic intent routing, escalation/fallback behavior, bounded conversation storage with TTL cleanup, and user-feedback capture for continuous quality monitoring.

Crew Call – AI Workforce Scheduling Engine

Node.js · LLM/Agent Workflows · MongoDB · Workforce Management APIs · Recommendation Logic · REST APIs

- Developed the engine layer that converts manager staffing requirements into eligible and ranked employee recommendations using workforce, scheduling, preference, and historical-assignment signals.
- Applied deterministic eligibility and business-rule checks around the AI layer so recommendations are validated before downstream scheduling actions are executed.
- Integrated Crew Call with enterprise workforce-management scheduling services and internal platform data flows for workforce lookup, schedule validation, and approved shift creation while keeping deterministic execution separate from the conversational AI layer.

Wikihire – Enterprise AI Recruitment & Requisition SaaS Platform
Node.js · GCP · Terraform · IAM · REST APIs · PostgreSQL · Docker

- Co-engineered backend routing and workflow services for an enterprise recruitment platform with multi-level approval flows and AI-assisted resume intelligence.
- Provisioned GCP resources with Terraform and managed IAM/service-account authentication across isolated environments for cloud deployment.
- Deployed Wikihire services on GCP and configured service-account token-based authentication for secure service-to-service access across environments.

Facial Recognition Attendance System – Major Airline Client
AWS Rekognition · Android · Geofencing · REST APIs · Backend APIs · Cost Modeling

- Built backend integration with Amazon Rekognition for employee face enrollment and real-time attendance verification, with configurable match-confidence thresholds.
- Integrated geofencing and attendance logging so check-in/check-out events are validated against office location and stored with employee, timestamp, and location context.

EDW Report Rationalization & AI-Assisted Mapping
LLMs · Excel · Enterprise Data Mapping · Power BI · ERP · Document Intelligence

- Built an AI-assisted workflow to classify legacy EDW reports as deprecated, existing-platform equivalent, ERP equivalent, new, or enhancement-required using enterprise reference documents.
- Mapped equivalent reports to internal/Power BI dashboards or ERP transaction codes and traced new/enhancement cases to supporting source documents for review evidence.
- Used LLM-assisted document comparison and semantic mapping to substantially reduce manual cross-file review and produce a consolidated, review-ready EDW mapping deliverable.

TECHNICAL SKILLS

- **AI & Machine Learning:** Machine Learning, Deep Learning, NLP, LLMs, RAG, AI Agents, Prompt Engineering, Semantic Search
- **Frameworks & Protocols:** LangChain, LangGraph, LlamaIndex, A2A, MCP, Scikit-learn, Pandas, NumPy
- **Backend & APIs:** Python, Node.js, JavaScript, REST APIs, MongoDB, PostgreSQL, Supabase, LLM API Integration
- **MLOps / LLMOps:** Model lifecycle, deployment strategies, monitoring and observability, evaluation, caching, pipeline design
- **Data & Analytics:** Snowflake, BigQuery, Power BI, Excel
- **Tools & Platforms:** OpenRouter, Groq, GCP, AWS Rekognition, Terraform, Docker, Git, Jupyter Notebook, Agentic IDEs

EDUCATION

Bachelor of Technology in Artificial Intelligence — Aug 2021 – Mar 2025
Dr. M.G.R. University, Chennai | 86%

Intermediate – MPC — Jun 2019 – May 2021
Viswa Sai Junior College | State Board | 91.2%

SSC — Jun 2018 – Apr 2019
GSEM High School | State Board of AP | 93%

CERTIFICATIONS & ACHIEVEMENTS

- Model Context Protocol (MCP) Developer Certification – Anthropic / Claude
- Claude with the Anthropic API – Anthropic
- Train and Evaluate Regression Models – Microsoft
- Train and Evaluate Deep Learning Models – Microsoft
- Cybersecurity in a Post-Quantum World – Workshop at SRM University, Chennai